Project Summary / Abstract

Agriculture is central to Uganda's economy, yet productivity remains constrained by limited access to timely agronomic guidance, climate variability, and an overstretched extension system. Farmers face increasing risks from frequent droughts, floods, and unpredictable rainfall, while existing advisory services are generalized and disconnected from localized data. At the same time, Uganda's expanding digital ecosystem offers an opportunity to deliver farm-specific, data-driven support at scale. This project proposes an AI-powered Agri-Climate Information and Digital Advisory (ACIDA) System to provide real-time, localized, and actionable climate and agronomic intelligence to farmers, extension agents, and institutions. The project objectives are to: (1) design, develop, and validate an AI–powered Agri-Climate Information and Digital Advisory (ACIDA) Systems, (2) establish an integrated geospatial reference database, (3) pilot and deploy the ACIDA system to deliver real-time agri-climate advisories, and (4) train and strengthen capacity of the farmers and stakeholders to effectively use ACIDA system (digital geospatial tools) and improve their decision making and resilience. A mixed-methods approach combining human-centered design, machine learning, geospatial analytics, IoT-enabled field data collection, and iterative piloting will be used. ACIDA will integrate multi-source datasets through APIs, process them via AI/ML models, and deliver personalized advisories through mobile apps, SMS/USSD, WhatsApp, and web dashboards. Capacity-building activities—including training-of-trainers, toolkits, and blended learning—will support uptake. Expected results include a functional ACIDA platform, an operational geospatial climate database, real-time advisory delivery in target areas, and strengthened farmer and institutional capacity. This novel project will enhance climate resilience, support adaptive farm planning, improve productivity, and contribute to Uganda's national climate commitments and sustainable food systems.

Description of the Problem, Logic, and Rationale of the Project

1 The Problem to be addressed by the proposed project

This project seeks to solve the problem of a lack of timely, localized, enterprise and farm-specific agri-climate information and services due to an overstretched agri-extension system and fragmented data sources. Although agricultural and climate data exist, they are not integrated and translated into useful products for local farmers. An AI–powered ACIDA decision-support system addresses these gaps by unifying multisource data, analyzing it in real time, and delivering personalized, predictive advisories that strengthen climate resilience and improve enterprise-specific and farm-level decision-making, including tailored weather forecasts and early warnings.

2 Logic and rationale of the project

The project is grounded in the recognition that Uganda's agriculture sector requires a shift from fragmented, low-information decision-making to an integrated, data-driven system capable of responding to climate and productivity challenges. Farmers and policymakers struggle to access timely, localized agronomic and climate information due to weak advisory systems, limited digital integration, and underutilization of existing datasets. Meanwhile, valuable data—such as weather forecasts, soil

maps, satellite imagery, crop models, and market intelligence—exists in multiple institutions but remains disconnected and difficult for end-users to interpret.

The logic of the project is that consolidating these diverse datasets into a unified ACIDA–powered platform enables real-time analysis, prediction, and delivery of actionable insights tailored to specific locations and farming systems. **The rationale is that the ACIDA** can translate complex data into simple recommendations, enhance climate risk preparedness, strengthen enterprise planning, and improve monitoring and reporting. This approach provides a scalable pathway to climate-smart, resilient, and productive agriculture in Uganda.

3 Proposed Solution: AI-powered Agri-Climate Information and Digital Advisory (ACIDA) Systems The proposed solution, ACIDA platform**,** is an integrated AI–powered agri-climate decision-support system designed to provide farmers, extension agents, and policymakers with real-time, localized, and actionable agronomic and climate intelligence. ACIDA tool consolidates diverse datasets—weather forecasts, satellite imagery, soil data, crop models, pest and disease alerts, market prices, and farmer-generated information—into a unified analytics engine. Using AI and machine learning, the system analyses these datasets to generate predictive agri-climate insights, enterprise and farm-specific advisories on optimal planting dates, input use, soil and water management, pest outbreaks, and climate adaptation strategies across all agroecological zones.

The proposed ACIDA tool features an intuitive dashboard that supports farmers, extensionists, and institutions with interactive mapping, crop monitoring tools, climate-risk alerts, and automated outputs for planning and ACIDA processes. Farmers access agri-climate advisories through mobile apps, SMS, IVR, WhatsApp, and community digital hubs, ensuring broad reach across literacy and connectivity levels. The solution enhances climate resilience, boosts productivity, and modernizes Uganda's agricultural decision-making ecosystem. It is organized into seven horizontal layers, representing core functional components of the tool (Figure 1).

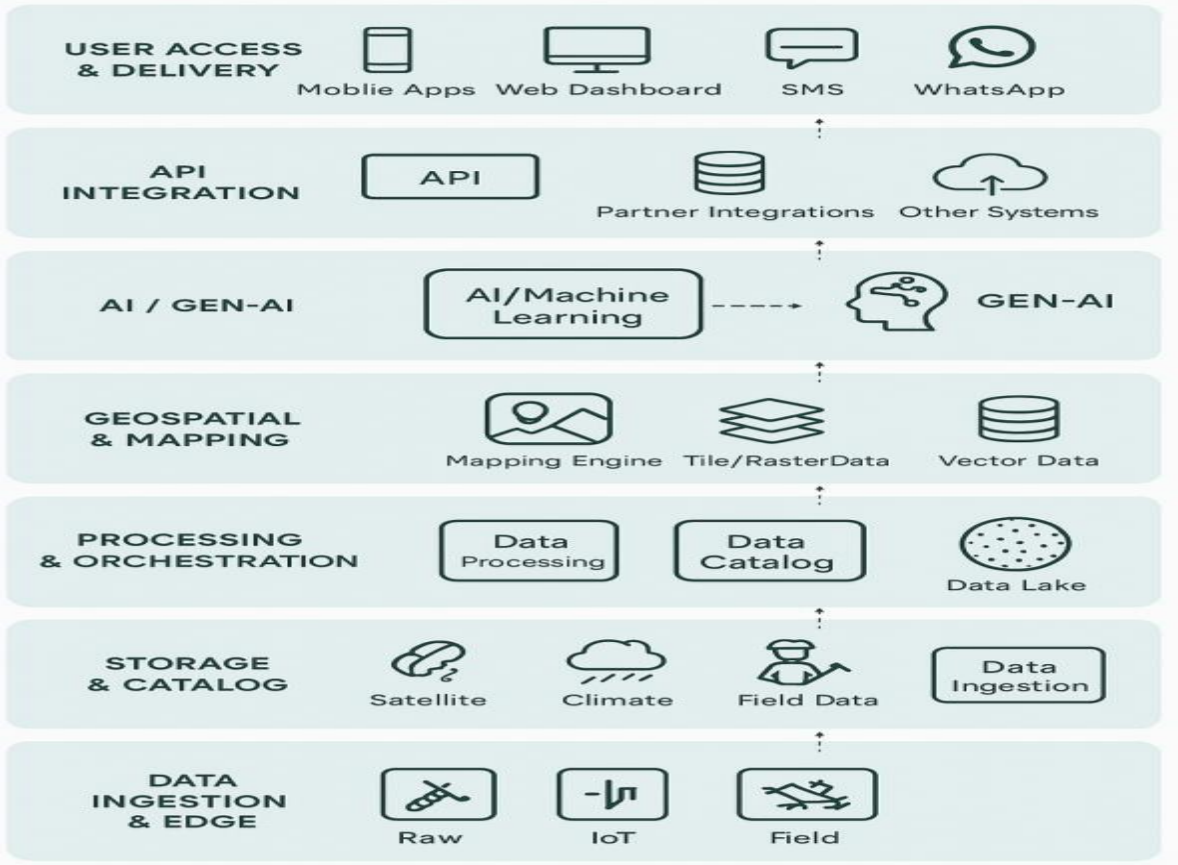


Figure 1: Schematic architecture of the new Agri-Climate Information and Digital Advisory (ACIDA) System

1) User Access & Delivery Layer: represents diverse channels end-users engage with ACIDA. These include mobile apps, a web dashboard, SMS, WhatsApp & other messaging platforms. By supporting phone interfaces, ACIDA guarantees inclusion of the digitally skilled and low-connectivity users, reinforcing advisory delivery.

2) API Integration Layer: features a central API gateway that links ACIDA with traditional digital ecosystems. It facilitates smooth data exchange with both national and international agric-tech databases, climate services, market information systems, satellite providers, and private agri-tech partners, enabling seamless interoperability.

3) ACIDA-AI Analytics Layer: Positioned at the system's core, this layer integrates machine learning models with the Generative AI capabilities. It performs predictive analytics, climate-risk assessment, automated content generation, and interpretation of complex climate and agronomic datasets, producing tailored advisory for users.

4) Geospatial & Mapping Layer: manages the geospatial data, including satellite imagery, raster tiles, and vector layers. It supports NDVI analysis, crop classification, land-use monitoring, and mapping of climate-risk hotspots across agroecosystems.

5) Processing and Orchestration Layer: this layer shows integrated components of the ACIDA system overseeing data processing pipelines, transformation, validation, and workflow orchestration. The agri-climate data catalogue is, therefore, included to ensure structured metadata tagging and organized data management, supported by a data lake for scalable storage for the end-users, mainly farmers and institutions.

6) Storage and Catalog Layer: this layer displays teh ACIDA system's categorized storage zones, including satellite data repositories, agri-climate datasets, field data, and ingested raw data. A scalable data lake and metadata catalogue enable efficient management of satellite data, climate datasets, field data observations, and partner-supplied information; ensuring a secure and scalable storage for diverse datasets.

7) Data Ingestion and Edge Layer: At the foundation of the ACIDA system, this layer captures the raw agri-climate data streams from the Internet of Things (IoT) sensors, weather stations, satellites, agroecology and farm-level inputs, forming the backbone of the system's analytics pipeline for national and global applications.

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4 Innovativeness of the proposed intervention

The proposed ACIDA tool introduces a transformative leap beyond conventional agri-climate advisory models by integrating generative artificial intelligence, real-time data analytics, and localized climate intelligence into a single interoperable platform. Unlike traditional extension services that rely on generic, infrequent, and often outdated recommendations, this system produces hyper-local, crop- and farm-specific guidance tailored to the farmers and institutions. Its ability to fuse satellite observations, IoT sensor data, historical climate patterns, biophysical models, and farmer-generated inputs enables dynamic prediction of climate risks (e.g., dry spells, floods), crop yield outcomes, soil health conditions, and pest and disease outbreaks. The platform's multilingual interfaces and voice assistant model democratize access to expert-quality advice, even for low-literate users. Moreover, AI continuously learns from feedback, strengthening accuracy over time. By bridging localized agri-climate information gaps, digitizing advisory ecosystems, and enabling inclusive, data-driven agricultural planning, the intervention establishes a new frontier in climate-smart, farmer-centric innovation.

Theory of Change (ToC)

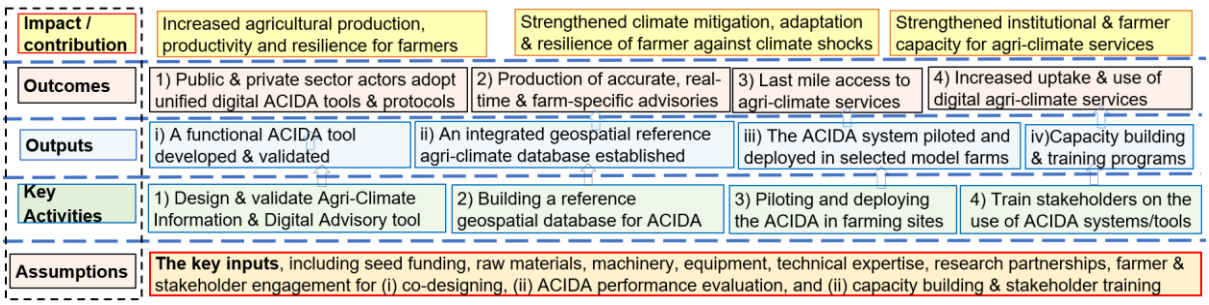


Figure 2: Schematic display of the *Theory of Change* for the proposed integrated digital ACIDA system

4.0 Goals: To increase agricultural productivity and resilience of farmers by facilitating creation, accessibility, and delivery of timely, accurate, enterprise and farm-specific agri-climate data and advisories in Uganda.

4.1 Objectives

1) To design, develop, and validate an AI–powered *Agri-Climate Information and Digital Advisory (ACIDA)* Systems, including analytical and utility models, ensuring accurate, scalable, and user-friendly functionality.

2) To establish an integrated geospatial reference database that integrates satellite, soil, climate, land-use, and crop profiles as backbones for generating accurate, timely, and farm-specific agri-climate data and advisories.

3) To pilot and deploy the ACIDA system to deliver real-time agri-climate information/advisories (seasonal forecasts, early warnings, and risk management actions to improve farmers' adaptive capacity and resilience.

4) To train and strengthen capacity of farmers and stakeholders with the skills and tools required to effectively use ACIDA system (digital geospatial tools) to improve their decision-making, adoption, and resilience.

Expected Outcomes

1) Farmers, public and private sector actors, and institutions adopt and use validated Agri-Climate Information and Digital Advisory (ACIDA) system to get accurate, timely, and site-specific agri-climate data/advisories.

2) An integrated geospatial reference agri-climate database established and functional, enabling the generation of agri-climate analytics and improving local and national capacity to produce accurate and timely advisories.

3) ACIDA piloted and deployed to deliver real-time agri-climate data and advisory services (forecasts, early warnings, and risk management) for enhanced preparedness, adaptive practices, and resilience of the farmers

4) Stakeholder capacities are strengthened across government, farmer organizations, and private sector actors, for increased uptake, effective use, and institutional integration of digital agri-climate and advisory solutions.

EXPECTED OUTPUTS

1. A fully functional and validated ACIDA, an agri-climate decision-support system developed for Uganda.

2. An integrated geospatial reference agri-climate database established for data analytics and services.

3. The ACIDA system was piloted and deployed in selected model farms and agroecological zones (AEZs).

4. Structured capacity building training programs and operational ACIDA toolkits for farmers and other actors.